

Accenture Technical Assessment Master Notes

Complete Technical Handout: Tier 1 Network Security, Cryptography, Attack Vectors & 50 Practiced MCQs

Placement Engineering

Network Security Tier 1

Ciphers & Algorithms

CIA Triad & Threats

50 High-Yield MCQs Included

1. AES — Advanced Encryption Standard

Definition: AES is a *symmetric-key block cipher* used globally to encrypt and decrypt sensitive data using the **exact same secret key**. It was established by the U.S. NIST in 2001 to replace DES because DES had become computationally insecure due to its small 56-bit key size.

How AES Works (Workflow)

Encryption: Plaintext → [AES Engine + Secret Key] → Ciphertext
Decryption: Ciphertext → [AES Engine + Same Secret Key] → Plaintext

Summary Rule: AES = Symmetric Encryption (Shared Secret Key)

AES Key Sizes, Block Size & Encryption Rounds

AES Variant	Key Size (Bits)	Rounds of Encryption	Block Size (Fixed)
AES-128	128 bits (16 bytes)	10 rounds	128 bits (16 bytes)
AES-192	192 bits (24 bytes)	12 rounds	128 bits (16 bytes)
AES-256	256 bits (32 bytes)	14 rounds	128 bits (16 bytes)



Critical MCQ Traps for AES:

- **Block Size is Invariable:** AES block size is **always 128 bits** regardless of whether AES-128, AES-192, or AES-256 is used.
- **Key Size ≠ Block Size:** In AES-256, the key size is 256 bits, but it processes data in 128-bit blocks over 14 rounds.
- **Block Cipher Mechanics:** Processes fixed 128-bit blocks of data simultaneously rather than continuous bit streams.

Key Advantages of AES

- **High Resistance:** Robust against differential, linear, and algebraic cryptanalysis.
- **Speed & Efficiency:** Hardware-accelerated (AES-NI instructions in modern CPUs), significantly faster than asymmetric ciphers.
- **Standardization:** De-facto worldwide standard for government, financial, and enterprise security.

Accenture MCQ Drill: Which of the following is a symmetric encryption algorithm?

- A. RSA | B. AES | C. ECC | D. Diffie-Hellman

✓ Correct Answer: B. AES (Uses a single shared key; RSA, ECC, and Diffie-Hellman are asymmetric/key-exchange).

2. DES — Data Encryption Standard

Definition: DES is an early *symmetric-key block cipher* developed by IBM in the 1970s. It operates on 64-bit plaintext blocks and executes **16 rounds** of a Feistel cipher structure.

DES Internal Structure & Flow

Plaintext → 64-bit Block → Initial Permutation (IP) → 16 Feistel Rounds → Final Permutation (FP / IP^{-1}) → Ciphertext

Critical Numerical Properties of DES

Property	DES Specification	Key Exam Takeaway
Block Size	64 bits	Processes 8 bytes at a time
Total Key Length	64 bits	Every 8th bit is discarded for parity verification
Effective Key Size	56 bits	Actual cryptographic security strength = 2^{56} possibilities
Number of Rounds	16 rounds	Uses 16 subkeys generated by key schedule

Why DES is Obsolete & Insecure: The 56-bit effective key space ($2^{56} \approx 7.2 \times 10^{16}$ keys) can be cracked via distributed brute-force in a matter of hours or minutes on modern computing hardware.

DES vs. AES Comparative Matrix

Feature	DES (Data Encryption Standard)	AES (Advanced Encryption Standard)
Cipher Type	Symmetric Block Cipher	Symmetric Block Cipher
Block Size	64 bits	128 bits
Effective Key Length	56 bits	128, 192, or 256 bits
Rounds	16 rounds	10, 12, or 14 rounds
Design Architecture	Feistel Network	Substitution-Permutation Network (SPN)
Security Status	Broken / Obsolete	Current International Standard (Secure)

3. RC4 & 4. RC5 (Stream vs. Block Ciphers)

RC4 (Rivest Cipher 4)

STREAM CIPHER **SYMMETRIC**

- Encrypts data byte-by-byte (or bit-by-bit) continuously by XORing plaintext with a pseudo-random keystream.
- Uses variable key sizes (40 to 2048 bits).
- Historically used in SSL/TLS 1.0, WEP, and WPA.
- **Status:** Deprecated and insecure due to statistical keystream biases (RFC 7465).

RC5 (Rivest Cipher 5)

BLOCK CIPHER **SYMMETRIC**

- Parametric symmetric block cipher designed by Ronald Rivest.
- **Configurable parameters:**
 - Word size: 16, 32, or 64 bits (block size = 2 × word size).
 - Number of rounds: 0 to 255.
 - Key size: 0 to 2040 bits.
- Features data-dependent bitwise rotations.

🔥 Master Memory Rule for Accenture:

- **RC4 = Stream Cipher** (Symmetric)
- **RC5 = Block Cipher** (Symmetric)
- **AES = Block Cipher** (Symmetric, 128-bit block)
- **DES = Block Cipher** (Symmetric, 64-bit block)

5. Symmetric vs. Asymmetric Cryptography

Symmetric Encryption (Single Shared Key): Fast and lightweight; perfect for bulk data. Both communicating parties must pre-share the secret key over a secure channel (leading to the *Key Distribution Problem*).

Sender: Plaintext → [Encrypt with Shared Secret Key] → Ciphertext
Receiver: Ciphertext → [Decrypt with Same Secret Key] → Plaintext

Asymmetric Encryption (Public-Private Key Pair): Uses two mathematically linked keys. The **Public Key** is freely distributed to everyone, while the **Private Key** is kept strictly confidential by the owner. Solves the key distribution issue and provides digital signatures.

Confidentiality Flow:

Sender → Encrypts using Receiver's PUBLIC Key → Ciphertext
Receiver → Decrypts using Receiver's PRIVATE Key → Plaintext

Digital Signature / Authentication Flow:

Signer → Signs/Encrypts hash using Signer's PRIVATE Key → Signature
Verifier → Verifies using Signer's PUBLIC Key → Verified Hash

Head-to-Head Comparison Matrix

Feature	Symmetric Encryption	Asymmetric Encryption
Number of Keys	1 shared secret key	2 keys: 1 Public + 1 Private key pair
Computational Speed	Extremely fast (100x to 1000x faster)	Slow (heavy modular exponentiation & elliptic curves)
Bulk Data Suitability	Ideal (large files, database disks, network payloads)	Inefficient for large payloads (used for small payloads/keys)
Core Algorithms	AES, DES, 3DES, RC4, RC5, ChaCha20	RSA, ECC (Elliptic Curve Cryptography), Diffie-Hellman, DSA
Primary Vulnerability / Issue	Secure key exchange / distribution problem	High CPU consumption and vulnerability to quantum Shor's algorithm

6. Encryption vs. Decryption Terminology

Term	Exact Technical Definition	Example Transformation
Plaintext	The original, human-readable data before cryptographic transformation.	"HELLO WORLD"
Ciphertext	The encrypted, unreadable output produced by a cipher.	"9f8a4e2b01c7d3"
Encryption	Process of converting Plaintext $\rightarrow$ Ciphertext to ensure Confidentiality.	$C = E_k(P)$
Decryption	Process of converting Ciphertext $\rightarrow$ Plaintext to recover information.	$P = D_k(C)$
Cryptographic Key	A string of bits fed into the mathematical algorithm to govern output.	256-bit binary/hex value

7. Cyber Threats, Social Engineering & Malware Taxonomies

DoS vs. DDoS Attacks

DoS (Denial of Service) - Launched from a single origin / machine. - Floods server resources (bandwidth, CPU, memory, socket connections) to take down services. CIA Pillar Targeted: AVAILABILITY - Easier to detect and mitigate via single-IP firewall rate-limiting or blackholing.	DDoS (Distributed Denial of Service) - Launched from thousands/millions of distributed bots (Botnet) under command of a botmaster. - Generates volumetric traffic (SYN flood, UDP amplification, HTTP flood). - Extremely difficult to block since malicious packets originate from globally dispersed authentic-looking IPs.
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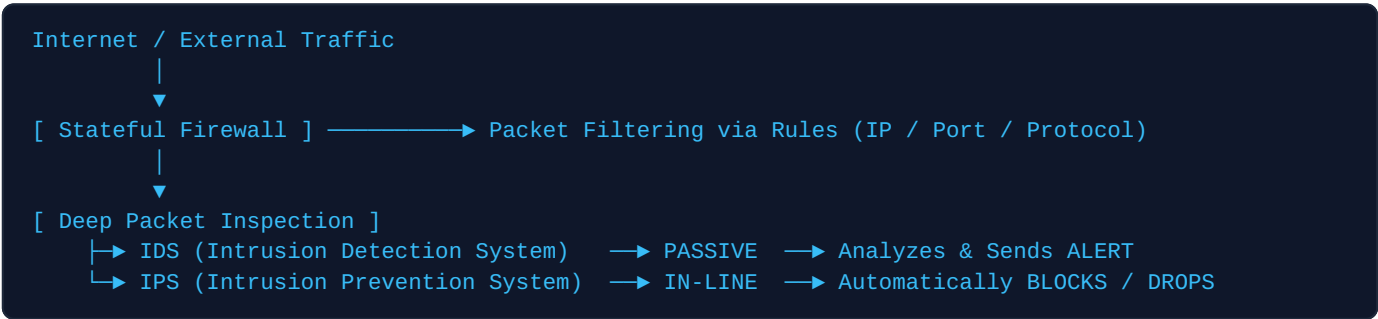
Malware Classification Matrix

Malware Type	Key Trigger / Defining Trait	Replication Behavior	Primary Objective
Ransomware	"Files encrypted & extortion/ ransom demanded"	Varies (often payloads dropped by trojans/exploits)	Financial extortion by locking user storage
Trojan Horse	"Disguised as legitimate, useful software"	Does NOT self-replicate	Creates backdoors, downloads rootkits, steals credentials
Worm	"Self-replicates across networks without host files"	Self-replicates automatically	Consumes network bandwidth and infects connected nodes
Virus	"Attaches to executable host files"	Replicates when infected host program is executed	Corrupts files, damages boot sectors, crashes system
Spyware / Keylogger	"Covert monitoring & background keystroke logging"	Non-replicating; installed stealthily	Harvests login credentials, credit cards, activity logs

Phishing vs. Spoofing

- **Phishing:** A *social engineering attack* exploiting human psychology/trust. Attackers craft fraudulent emails, SMS (Smishing), or fake login portals to trick users into divulging credentials or OTPs.
- **Spoofing:** A *technical mechanism of falsifying identity metadata*. Examples include forging the sender in the email header (*Email Spoofing*) or forging the source IP address in packet headers (*IP Spoofing*).
- *Relationship:* Phishing frequently uses email spoofing as a vehicle to appear authentic.

8. Network Defense Infrastructure: Firewall, IDS, IPS & UTM



IDS vs. IPS Comparison

Feature	IDS (Intrusion Detection System)	IPS (Intrusion Prevention System)
Deployment Mode	Out-of-band / Passive (monitoring span port / tap)	In-line directly in the active network traffic path
Action on Malicious Packet	Generates log alerts, notifies administrator; allows packet to pass	Actively drops packets , resets TCP connection, blocks source IP
Impact on Latency	Zero network latency impact	Slight latency overhead due to real-time deep packet filtering
Accenture Memory Hook	IDS = DETECT & ALERT	IPS = DETECT & PREVENT/BLOCK

UTM (Unified Threat Management)

Definition: UTM is an all-in-one consolidated security appliance or cloud solution that combines multiple network security technologies into a unified management architecture:

UTM = Firewall + IDS/IPS + Gateway Antivirus + Web Filtering + Anti-Spam + VPN Gateway

Exam Tip: A firewall is just a single component within a comprehensive UTM ecosystem.

9. CIA Triad, AAA Security Framework & Access Control

Security Pillar	Core Question / Meaning	Primary Safeguard Mechanism	Primary Threat
Confidentiality	Data is readable only by authorized individuals	Symmetric & Asymmetric Encryption, ACLs	Eavesdropping, Snooping, Data Breaches
Integrity	Data remains unmodified, authentic, and unaltered	Cryptographic Hash Functions (SHA-256), Digital Signatures	Man-in-the-Middle (MitM), Tampering
Availability	Systems and data are accessible 24/7 to authorized users	Redundancy, Load Balancers, Fault Tolerance	DoS, DDoS, Ransomware Lockouts
Authentication	"Who are you?" — Identity verification	Passwords, Multi-Factor Authentication (MFA), Biometrics	Identity Spoofing, Credential Stuffing
Authorization	"What are you permitted to do?" — Permissions	Role-Based Access Control (RBAC), Access Control Lists	Privilege Escalation

Part 1: 25 Foundational Accenture Practice MCQs

Q1. AES is a _____ encryption algorithm.

A. Asymmetric | B. Symmetric | C. Hashing | D. Digital signature

✓ **Answer: B (AES is a symmetric-key block cipher using a single shared secret).**

Q2. Which is the block size of AES?

A. 64 bits | B. 128 bits | C. 192 bits | D. 256 bits

✓ **Answer: B (AES block size is permanently fixed at 128 bits across all key variations).**

Q3. How many rounds does AES-128 perform?

A. 8 | B. 10 | C. 12 | D. 14

✓ **Answer: B (AES-128 performs 10 rounds; AES-192 does 12; AES-256 does 14).**

Q4. How many rounds does AES-256 perform?

A. 10 | B. 12 | C. 14 | D. 16

✓ **Answer: C (AES-256 performs 14 transformation rounds).**

Q5. What is the effective key size of DES?

A. 32 bits | B. 56 bits | C. 64 bits | D. 128 bits

✓ **Answer: B (64 bits total, but 8 parity bits leave 56 bits effective cryptographic strength).**

Q6. DES performs how many encryption rounds?

A. 8 | B. 10 | C. 16 | D. 32

✓ **Answer: C (DES Feistel network executes 16 identical encryption rounds).**

Q7. Which of the following is a stream cipher?

A. AES | B. DES | C. RC4 | D. RSA

✓ **Answer: C (RC4 encrypts continuous streams of bytes; AES and DES are block ciphers).**

Q8. Which statement about RC5 is correct?

A. It is an asymmetric algorithm | B. It is a stream cipher | C. It is a symmetric block cipher | D. It is a hashing algorithm

✓ **Answer: C (RC5 is a symmetric block cipher with parameterized block and key sizes).**

Q9. Which algorithm uses a public key and a private key?

A. AES | B. DES | C. RSA | D. RC4

✓ **Answer: C (RSA is asymmetric, using a mathematically linked key pair).**

Q10. Which encryption method is generally faster for encrypting large amounts of data?

A. Symmetric encryption | B. Asymmetric encryption | C. Hashing | D. Digital signature

✓ **Answer: A (Symmetric ciphers have negligible CPU overhead compared to asymmetric math).**

Q11. What is the main purpose of encryption?

A. Increase network speed | B. Convert ciphertext into plaintext | C. Protect data from unauthorized access | D. Detect viruses

✓ **Answer: C (Encryption preserves data confidentiality against unauthorized inspection).**

Q12. A company server becomes unavailable because an attacker sends a huge number of requests. This is an example of:

A. Phishing | B. DoS | C. Spoofing | D. Trojan

✓ **Answer: B (Denial of Service aims directly at exhausting availability).**

Q13. Thousands of compromised computers simultaneously attack a website. This is:

A. DoS | B. DDoS | C. Phishing | D. Brute force

✓ **Answer: B (Distributed DoS coordinates traffic across a distributed botnet).**

Q14. Which attack primarily exploits human trust to obtain passwords or sensitive information?

A. Phishing | B. DDoS | C. Buffer overflow | D. Packet sniffing

✓ **Answer: A (Phishing targets social vulnerabilities rather than software bugs).**

Q15. An attacker creates an email that appears to come from a legitimate bank, although it actually came from the attacker. This is:

A. Phishing only | B. Spoofing | C. Ransomware | D. DoS

✓ **Answer: B (Falsifying the origin identifier of a communication is spoofing).**

Q16. A malicious program encrypts a user's files and demands money to restore access. This is:

A. Trojan | B. Spyware | C. Ransomware | D. Worm

✓ **Answer: C (File locking + payment demand = Ransomware).**

Q17. Malware disguised as legitimate software is called:

A. Trojan | B. Ransomware | C. Worm | D. Firewall

✓ **Answer: A (Trojan masquerades as benign utilities to induce manual installation).**

Q18. Which malware is known for self-replicating across networks without necessarily attaching itself to another program?

A. Trojan | B. Worm | C. Ransomware | D. Spyware

✓ **Answer: B (Worms propagate stand-alone across network ports without host files).**

Q19. What is the primary function of an IDS?

A. Encrypt network traffic | B. Detect suspicious activity and generate alerts | C. Automatically replace infected files | D. Assign IP addresses

✓ **Answer: B (IDS is passive detection and alerting; it does not in-line drop traffic).**

Q20. Which security system can detect and actively block malicious traffic?

A. IDS | B. IPS | C. DNS | D. DHCP

✓ **Answer: B (IPS operates in-line to actively block/prevent cyber threats).**

Q21. UTM stands for:

A. Unified Traffic Management | B. Universal Threat Monitoring | C. Unified Threat Management | D. User Threat Management

✓ Answer: C (Unified Threat Management).

Q22. Which statement best describes UTM?

A. It only provides encryption | B. It combines multiple security functions into one solution | C. It only detects viruses | D. It is a replacement for TCP

✓ Answer: B (Consolidates firewall, IPS, AV, content filtering in one architecture).

Q23. Which of the following is NOT a symmetric encryption algorithm?

A. AES | B. DES | C. RC4 | D. RSA

✓ Answer: D (RSA is asymmetric).

Q24. Which of the following pairs is correctly matched?

A. AES — Asymmetric | B. RSA — Symmetric | C. RC4 — Stream cipher | D. DES — Hash function

✓ Answer: C (RC4 is a symmetric stream cipher).

Q25. Which CIA security property is primarily affected by a DoS attack?

A. Confidentiality | B. Integrity | C. Availability | D. Authentication

✓ Answer: C (Denial of Service denies authorized access, violating Availability).

**Part 2: 25 Scenario-Based Placement Exam MCQs****Q1. A company's employees receive an email that appears to be from the IT department. It asks them to click a link and enter their username and password on a fake login page. What attack is this?**

A. DDoS | B. Phishing | C. Spoofing only | D. Ransomware

✓ Answer: B (Phishing — deceptive communication crafted to steal credentials).

Q2. A bank's website suddenly receives millions of requests from thousands of different compromised computers. The server becomes unavailable to legitimate customers. What is happening?

A. DoS | B. DDoS | C. Phishing | D. Trojan attack

✓ Answer: B (DDoS — simultaneous volumetric flooding via thousands of distributed machines).

Q3. An attacker sends a network packet with a forged source IP address to make the packet appear as if it came from a trusted machine. Which technique is being used?

A. Phishing | B. IP spoofing | C. Ransomware | D. DDoS

✓ **Answer: B (IP Spoofing — header forgery of the source IP address).**

Q4. A company wants to protect confidential customer data stored in its database. It needs a fast encryption algorithm for encrypting a large amount of data. Which is the most appropriate choice?

A. RSA | B. AES | C. Digital signature | D. Hashing

✓ **Answer: B (AES — industry standard symmetric cipher providing high performance on bulk data).**

Q5. A developer receives an encrypted message intended only for them. The sender uses the developer's public key to encrypt the message. Which key should the developer use to decrypt it?

A. Sender's public key | B. Sender's private key | C. Developer's public key | D. Developer's private key

✓ **Answer: D (Developer's private key — only the recipient's private key can decrypt data sealed with their public key).**

Q6. A company discovers malware that secretly encrypts thousands of employee files. The attackers then demand cryptocurrency to provide the decryption key. What type of malware is this?

A. Trojan | B. Worm | C. Ransomware | D. Spyware

✓ **Answer: C (Ransomware — malicious encryption coupled with extortion).**

Q7. An employee downloads a software application that appears to be a legitimate PDF reader. After installation, the software secretly installs malicious code. What is the best classification?

A. Trojan | B. DDoS | C. Phishing | D. Firewall

✓ **Answer: A (Trojan — malicious utility masquerading as a benign application).**

Q8. A security administrator wants a system that monitors network traffic and alerts the administrator whenever suspicious activity is detected, but does not automatically block the traffic. Which should be deployed?

A. IPS | B. IDS | C. UTM | D. VPN

✓ **Answer: B (IDS — Intrusion Detection System is passive, alerting without blocking).**

Q9. A company wants a security system that detects malicious network traffic and automatically blocks it before it reaches internal systems. Which technology is most appropriate?

A. IDS | B. IPS | C. DNS | D. DHCP

✓ **Answer: B (IPS — Intrusion Prevention System is in-line and automatically drops threats).**

Q10. A small organization wants one security solution that provides firewall, intrusion prevention, antivirus, and web filtering capabilities. Which solution best fits the requirement?

A. UTM | B. DNS | C. AES | D. IDS only

✓ **Answer: A (UTM — Unified Threat Management bundles all perimeter security functions).**

Q11. A company uses AES-256 to encrypt sensitive files. Which statement is correct?

A. AES-256 uses a 256-bit block size | B. AES-256 uses a 256-bit key and 128-bit block size | C. AES-256 is asymmetric
| D. AES-256 uses a 56-bit key

✓ **Answer: B (Key length = 256 bits, but AES block size remains 128 bits across all iterations).**

Q12. A security engineer is analyzing an old application and discovers that it uses DES for encryption. Why is replacing DES with AES recommended?

A. DES is asymmetric | B. DES has a small effective key size and is vulnerable to brute-force attacks | C. AES does not use encryption keys | D. DES cannot encrypt data

✓ **Answer: B (DES 56-bit key is easily crackable via modern brute-force hardware).**

Q13. A company needs encryption for a large video file. The engineer chooses RSA directly because it provides strong security. What is the main problem with this approach?

A. RSA is generally slower and inefficient for bulk data encryption | B. RSA cannot use keys | C. RSA is a stream cipher |
D. RSA provides no security

✓ **Answer: A (Asymmetric RSA is computationally heavy; symmetric AES is required for bulk data).**

Q14. An attacker sends millions of requests to a server from a single attacking machine, exhausting the server's resources. Which attack is most appropriate?

A. DDoS | B. DoS | C. Phishing | D. Spoofing

✓ **Answer: B (DoS originates from a single attacking machine).**

Q15. A security analyst sees that a malicious email appears to come from ceo@company.com, but the actual sender is an attacker. Which concept explains the attacker making the email appear to originate from another source?

A. Spoofing | B. Ransomware | C. Encryption | D. IDS

✓ **Answer: A (Email spoofing falsifies the SMTP RFC 5322 header address).**

Q16. An attacker sends an email containing a fake Microsoft login page. The attacker also makes the email appear to come from Microsoft's official address. Which combination is being used?

A. Phishing + spoofing | B. DDoS + ransomware | C. Trojan + worm | D. IDS + IPS

✓ **Answer: A (Fake login page = Phishing; forged sender address = Spoofing).**

Q17. A network administrator wants to protect a server from unauthorized network traffic by allowing only traffic matching predefined rules based on IP addresses, ports, and protocols. What should be used?

A. Firewall | B. AES | C. Trojan | D. RSA

✓ **Answer: A (Firewalls enforce rule-based traffic filtering based on L3/L4 attributes).**

Q18. A security system detects an attack and immediately drops the malicious packets instead of merely generating an alert. Which technology is most likely being used?

A. IDS | B. IPS | C. DNS | D. VPN

✓ **Answer: B (IPS actively drops packets).**

Q19. A company wants to ensure that only authorized employees can access the payroll database. Which security concept primarily addresses "who is allowed to access what?"

A. Authentication | B. Authorization | C. Encryption | D. Availability

✓ **Answer: B (Authorization governs permissions and privilege levels).**

Q20. An employee enters their username and password into a company system. The system verifies that the credentials belong to that employee. This process is:

A. Authorization | B. Authentication | C. Spoofing | D. Encryption

✓ **Answer: B (Authentication proves identity: "Who are you?").**

Q21. A company wants to ensure that an attacker cannot secretly modify financial transaction data while it is being transmitted. Which CIA property is primarily concerned?

A. Confidentiality | B. Integrity | C. Availability | D. Authentication

✓ **Answer: B (Integrity prevents and detects unauthorized tampering or alteration).**

Q22. A company's online banking service must remain accessible to customers 24/7. An attacker floods the service with traffic, causing legitimate customers to lose access. Which CIA property is primarily affected?

A. Confidentiality | B. Integrity | C. Availability | D. Authorization

✓ **Answer: C (Resource exhaustion directly denies service Availability).**

Q23. A company uses the same secret key to encrypt and decrypt a large database. Which type of cryptography is being used?

A. Asymmetric | B. Symmetric | C. Hashing | D. Digital signature

✓ **Answer: B (Same secret key for both operations = Symmetric Cryptography).**

Q24. A security engineer needs a stream cipher for a legacy system and is given these options: AES, DES, RC4, RSA. Which should be selected based purely on the algorithm type?

A. AES | B. DES | C. RC4 | D. RSA

✓ **Answer: C (RC4 is the classic symmetric stream cipher).**

Q25. A company's security architecture has the following requirements: Detect suspicious traffic, automatically block detected attacks, provide firewall functionality, provide antivirus protection, provide web filtering. Which option best satisfies all requirements?

A. IDS | B. IPS only | C. UTM | D. AES

✓ **Answer: C (Unified Threat Management combines all these perimeter defenses into one system).**

Quick-Revision Summary Matrix

Topic	Key Takeaways & Formulas	Accenture Exam Trap
AES	Block = 128-bit; Keys = 128 (10 rounds), 192 (12 rounds), 256 (14 rounds)	Do not confuse key size with block size. Block size is never 256 bits.
DES	Block = 64-bit; Rounds = 16; Effective Key = 56 bits (8 bits parity)	Total key = 64 bits, but effective cryptographic strength = 56 bits.
RC4 vs RC5	RC4 = Stream Cipher; RC5 = Block Cipher (variable parameters)	RC4 processes byte streams; RC5 processes blocks.
Symmetric vs Asymmetric	Symmetric = 1 key (AES, DES, RC4); Asymmetric = 2 keys (RSA, ECC)	Symmetric is fast for bulk data; Asymmetric is slow, used for key exchange.
IDS vs IPS	IDS = Passive (Alerts); IPS = In-line Active (Blocks / Drops)	IDS does not drop packets; IPS automatically drops malicious traffic.
DoS vs DDoS	DoS = Single origin machine; DDoS = Botnet (distributed machines)	Both violate the Availability pillar of the CIA Triad.
Authentication vs Authorization	AuthN = Identity ("Who you are"); AuthZ = Permissions ("What you can do")	Passwords authenticate; Role permissions authorize.
UTM	Unified Threat Management (Firewall + IPS + AV + Web Filter in one box)	A firewall is a single module inside UTM, not a synonym.